Ch. 30 HW Solutions

Principles of Macroeconomics

March 7, 2026

Quick Quiz

1. (b) – Fiat money is money that only has value because we've collectively decided it has value.
2. (c) – Paying with a credit card isn't actually paying; it's promising to pay later. A promise to pay later isn't a reliable medium of exchange, unit of account, or store of value.
3. (a) – The Fed was established by Congress.
4. (c) – If it buys bonds, it pays the bond owners. When it pays the bond owners, it can introduce new money into the economy.
5. (c) – If the bank doesn't loan out some of the deposits, then the money supply is unchanged because \$100 of currency is converted into \$100 of deposits. If the bank does loan out some of the deposits, the money supply increases because \$100 of currency is converted into more than \$100 of deposits. That's because fractional-reserve banking creates money.
6. (d) –
$$\$120 \cdot \frac{1}{0.25} = \$480$$
7. (c) – The percent change in bank capital is $5 \cdot -10\% = -50\%$. A 50% decrease in \$200 leaves \$100.

Questions for Review

1. Money is a medium of exchange and a unit of account.
2. Commodity money has intrinsic value. Fiat money does not. Fiat money only has value because we've decreed it has value. We use fiat money
3. Demand deposits are deposits that you can withdraw whenever you want. Typically, they're checking account deposits. They're included in the money stock because you can use a debit card to pay with deposits just as easily as you can use cash.

4. The Fed sets monetary policy. The members of the Fed who vote on monetary policy are split between Presidential appointees and regionally appointed officials.

5. It buys government bonds.

6. If they lend out some of their deposits, they can charge interest on it, and earn a profit. Reserves and the money supply are related as follows:

$$S_M = C + D$$
$$D = \text{res} \cdot MM = \text{res} \cdot \frac{1}{\text{res ratio}}$$
$$S_M = C + \text{res} \cdot \frac{1}{\text{res ratio}}$$

Problems and Applications

1. (a) Money – Pennies are currency. They're accepted as a medium of exchange. Prices are denominated in dollars and cents, so they're a unit of account. And they store value because you can save a penny today and spend it in the future.

   (b) No, Mexican pesos are not a medium of exchange in the U.S. because they're not regularly accepted as payment. Similarly they are not the unit of account because prices are quoted in dollars. That said, they are a store of value because Mexican pesos are assets that can always be exchanged for U.S. dollars.

   (c) No, a Picasso painting is not the unit of account in the U.S. because prices aren't quoted in Picasso paintings. Similarly, it's not a medium of exchange because Picasso paintings aren't regularly used to buy things. It is a store of value because it will have value in the future.

   (d) No, a plastic credit card is not a medium of exchange because it is not a payment method; it's a method of deferring payment. A credit card payment is a store of value – although not a reliable one – because it is a promise to pay later. It's not a unit of account because prices are denoted in today's dollars, not future dollar payments.

2. (a) Increase – the Fed buys bonds with newly created dollars, so it adds more money to the money supply.

   (b) Increase – a lower reserve requirement lowers the reserve ratio. That increases the money multiplier which in turn increases money creation by banks.

   (c) Decrease – a higher interest rate on reserves incentivizes banks to hold more reserves. More reserves leads to a higher reserve ratio and therefore a lower money multiplier. A lower money multiplier decreases money creation by banks.

(d) Decrease – the Fed takes the money it receives out of circulation, so the money supply decreases.

(e) Increase – when people deposit their currency at banks, banks then make loans using the deposits. The process of taking deposits and making loans generates money, so the money supply increases.

(f) Decrease – an increase in reserves leads to an increase in the reserve ratio. An increase in the reserve ratio decreases the money multiplier. When the money multiplier decreases, banks create less money, so the money supply shrinks.

(g) Decrease – the Fed increases its target for the federal funds rate by increasing the interest it pays on reserves. When the Fed increases the interest it pays on reserves, the money supply decreases for the reasons in part (c).

4. (a) If deposits are \$250 mil, then reserves must be $0.1 \cdot \$250 = \$25$. That leaves $\$250 - \$25 = \$225$ for loans.

Assets	Liabilities
Reserves: \$25 mil	Deposits: \$250 mil
Loans: \$225 mil	

(b) After the \$10 mil withdrawal, deposits are \$240 mil and reserves are \$15 mil. The bank needs reserves to be \$24 mil, so it must call in \$9 million of loans. That leaves \$216 mil in loans outstanding.

Assets	Liabilities
Reserves: \$24 mil	Deposits: \$240 mil
Loans: \$216 mil	

(c) In order to pay back BSB, people might have to withdraw money from other banks. In that scenario, other banks will need to restore their reserve ratios as well, so they might have to reduce the amount of loans they have outstanding as well. Overall, if Daddy Warbucks keeps his \$10 million in cash, then reserves will decrease by \$10 million in the banking system as a whole. If reserves decrease by \$10 million, then deposits decrease by more than \$10 million due to the money multiplier. As a result, the money supply decreases.

(d) Banks can't force people to pay back their loans ahead of schedule, so BSB might not be able to reduce the amount of loans it has outstanding immediately. Instead it could try to increase reserves by attracting more deposits, borrowing from the Fed, or borrowing from other sources.

5.

$$\text{Money Multiplier} = \frac{1}{\text{Reserve Ratio}} = \frac{1}{0.1} = 10$$

$$\Delta\text{Deposits} = \Delta\text{Reserves} \cdot (\text{Money Multiplier}) = \$100 \cdot 10 = \$1,000$$

$$\Delta\text{Money Supply} = \Delta\text{Currency} + \Delta\text{Deposits} = -\$100 + \$1,000 = \$900$$

6. (a)

Assets	Liabilities
Reserves: $\$800 \cdot \frac{1}{8} = \$100$	Deposits: \$800
Loans: \$800 + \$200 - \$100 = \$900	Capital: \$200

(b)

$$\text{Leverage Ratio} = \frac{\text{Assets}}{\text{Capital}} = \frac{\$100 + \$900}{\$200} = 5$$

(c)

Assets	Liabilities
Reserves: $\$800 \cdot \frac{1}{8} = \$100$	Deposits: \$800
Loans: $\$900 \cdot 0.9 = \$810$	Capital: $\$100 + \$810 - \$800 = \$110$

(d) The banks total assets declined by $90/1,000 = 9\%$. The bank's capital declined by $90/200 = 45\%$. The decline in capital is larger because the bank's leverage amplifies their gains and losses.

7. If the government buys \$10 million of government bonds and people hold all \$10 million in currency, then the money supply will increase by \$10 million dollars. That's the smallest possible increase in the money supply because banks can't generate any additional money if people don't make any additional deposits.

   If people deposit the original \$10 million, and if people continually deposit the money that banks loan out from that original \$10 million, then reserves will increase by \$10 million. If banks don't hold excess reserves, then the reserve ratio will be 10%, so the money multiplier will be $1/0.1 = 10$. That means deposits will increase by $10 \cdot \$10\text{million} = \$100$ million. That's the largest possible increase in the money supply because that's the most additional money that banks can generate.

8. The Fed buying \$2,000 worth of bonds will increase the money supply more. If people don't deposit the new currency, then the money supply will increase by \$2,000. If people do deposit the new currency and if people continually deposit the money that banks loan out from those new deposits, reserves will increase by \$2,000. If reserves increase by \$2,000, then deposits will increase by $\$2,000 \cdot \frac{1}{0.05} = \$40,000$, so the money supply will increase by \$40,000.

   If someone deposits \$2,000 worth of currency and people don't deposit the money that banks loan out from those new deposits, the money supply doesn't change. Deposits increase by \$2,000, but currency decreases by \$2,000. If people continually deposit the money that banks loan out from

those new deposits, reserves increase by \$2,000, and deposits increase by \$40,000 for the same reason as above. That said, currency still decreases by \$2,000, so the money supply only increases by $-\$2,000 + \$40,000 = \$38,000$.